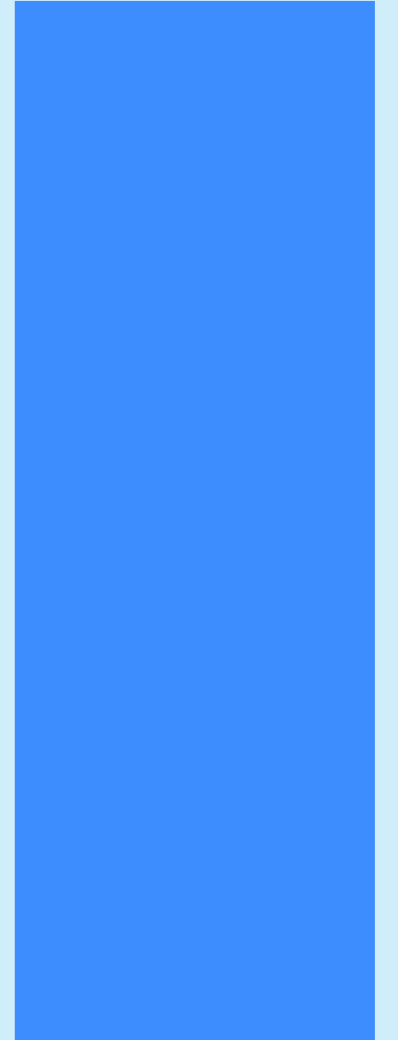


Skin Cancer Risk Intelligence

Dermalife Oncology & Research Institute (DORI)

Prepared by Tsitsi Maxine Ndudzo | PostgreSQL analysis | July 2026



DORI uses data to strengthen early detection

The organisation

Dermalife Oncology & Research Institute is a specialised healthcare organisation focused on clinical research, advanced diagnostics and population health analytics.

Its mission is to reduce skin-cancer mortality, improve early-stage diagnosis and support personalised, evidence-based care.

1,088

patient and lesion records analysed

4

risk dimensions: demographic, lesion, environment and lifestyle

SQL

joins, filtering, aggregation and comparative risk analysis

Rising late-stage diagnoses require a clearer risk framework

01

WHO IS AT RISK?

Identify age, sex and previous-history patterns linked to diagnosis demand.

02

WHAT SIGNALS DANGER?

Find symptoms, growth and lesion characteristics that can support earlier escalation.

03

WHERE TO INTERVENE?

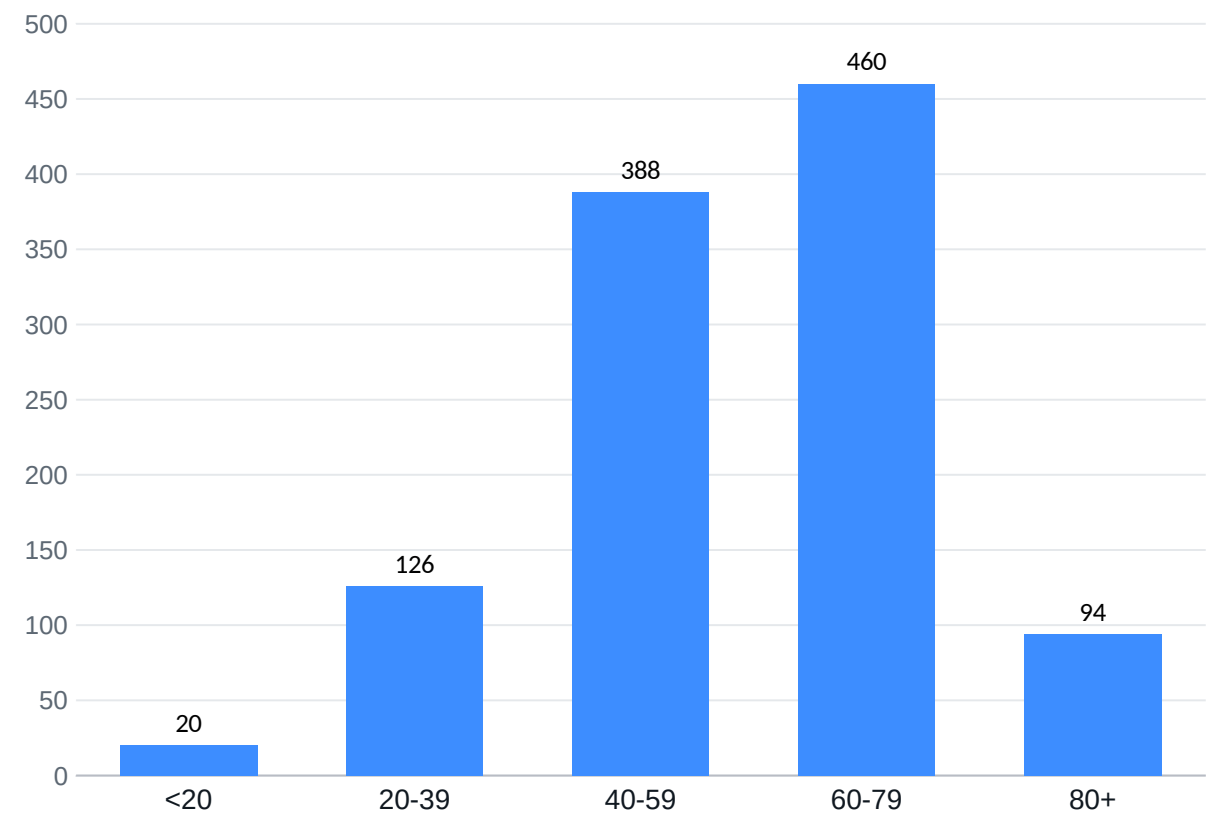
Compare body regions and access to water and sewage with diagnosis severity.

04

WHICH HABITS MATTER?

Assess smoking, alcohol and pesticide exposure without treating association as causation.

Ages 60-79 form the largest diagnosis group



66.7%

Male diagnoses
726 records

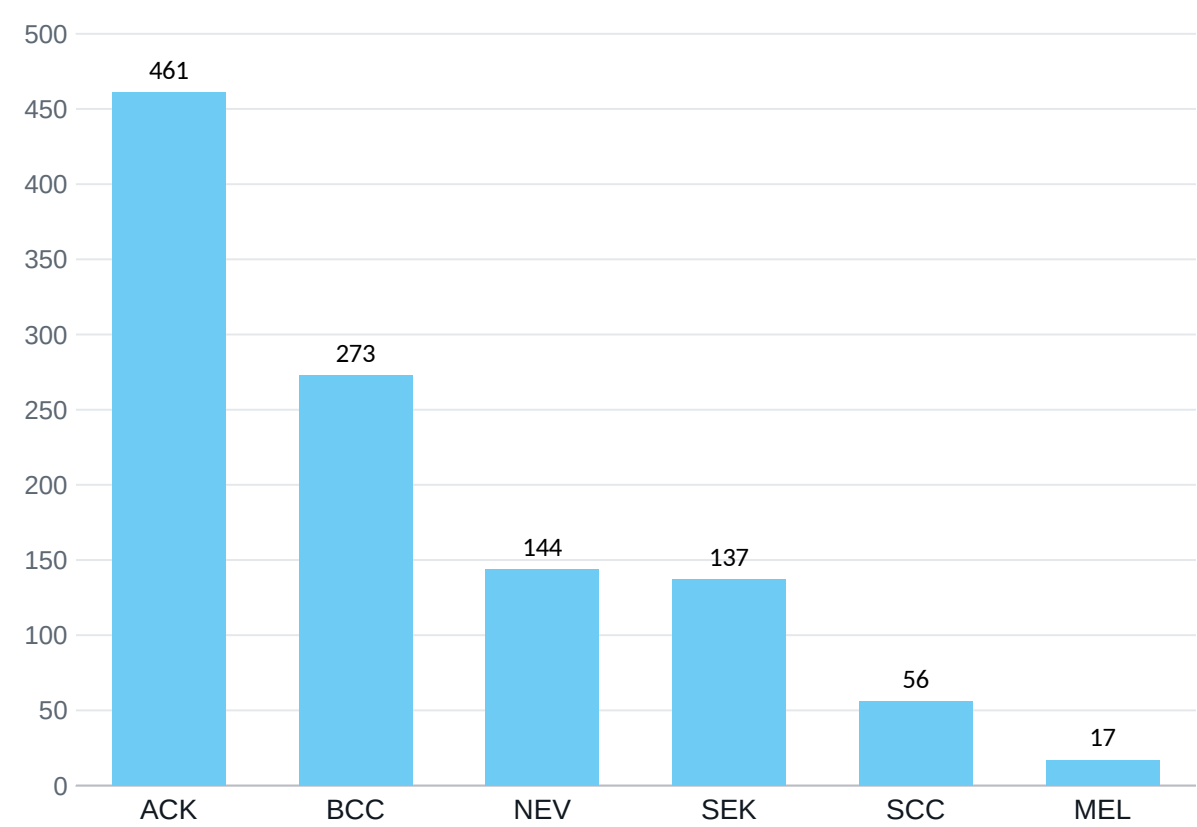
20.6%

Previous skin-cancer
history
224 patients

What this means

Prioritise recall and education for older adults, while reviewing why men contribute twice as many records as women.

ACK is most frequent; melanoma lesions are largest



670

itching records - the most common symptom

458

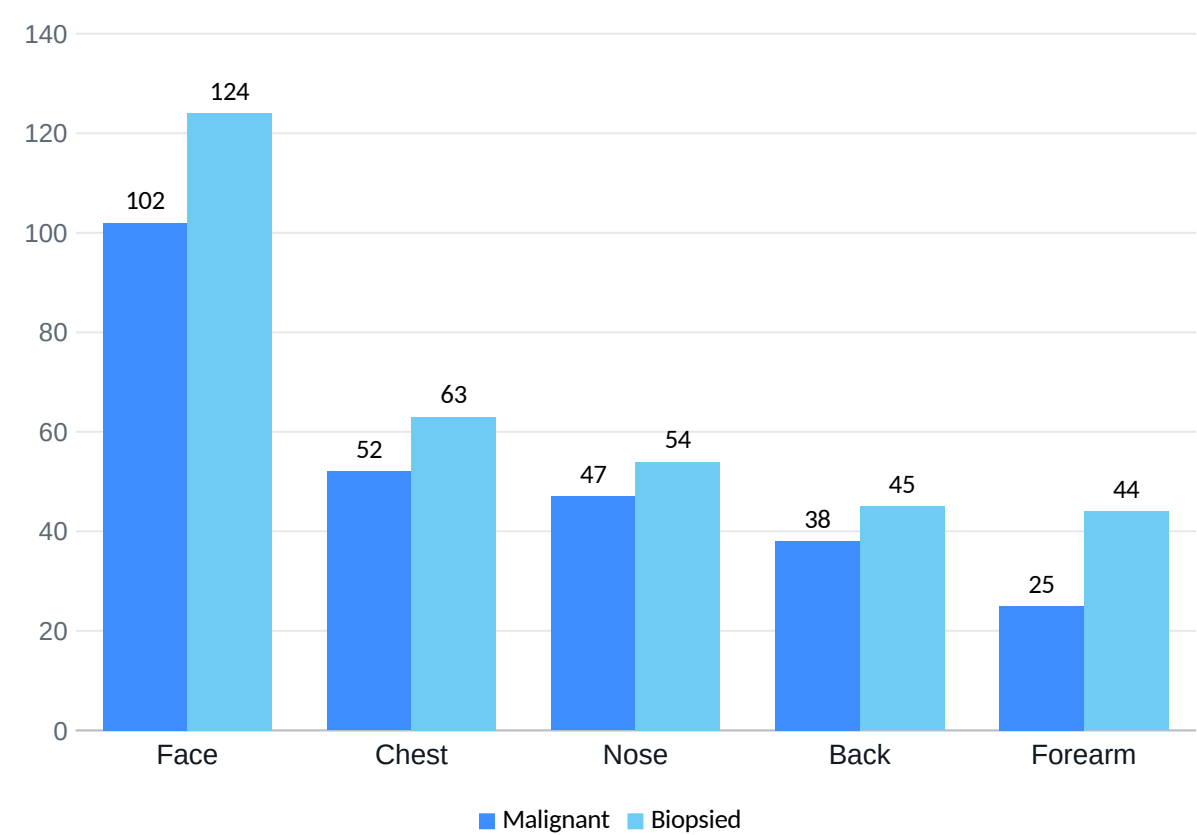
lesions biopsied (42.1% of all lesions)

14.09

average melanoma diameter - highest of all diagnoses

Abbreviations: ACK actinic keratosis; BCC basal cell carcinoma; MEL melanoma; SCC squamous cell carcinoma; NEV nevus; SEK seborrheic keratosis.

The face is the main clinical workload hotspot



278 face lesions - highest overall

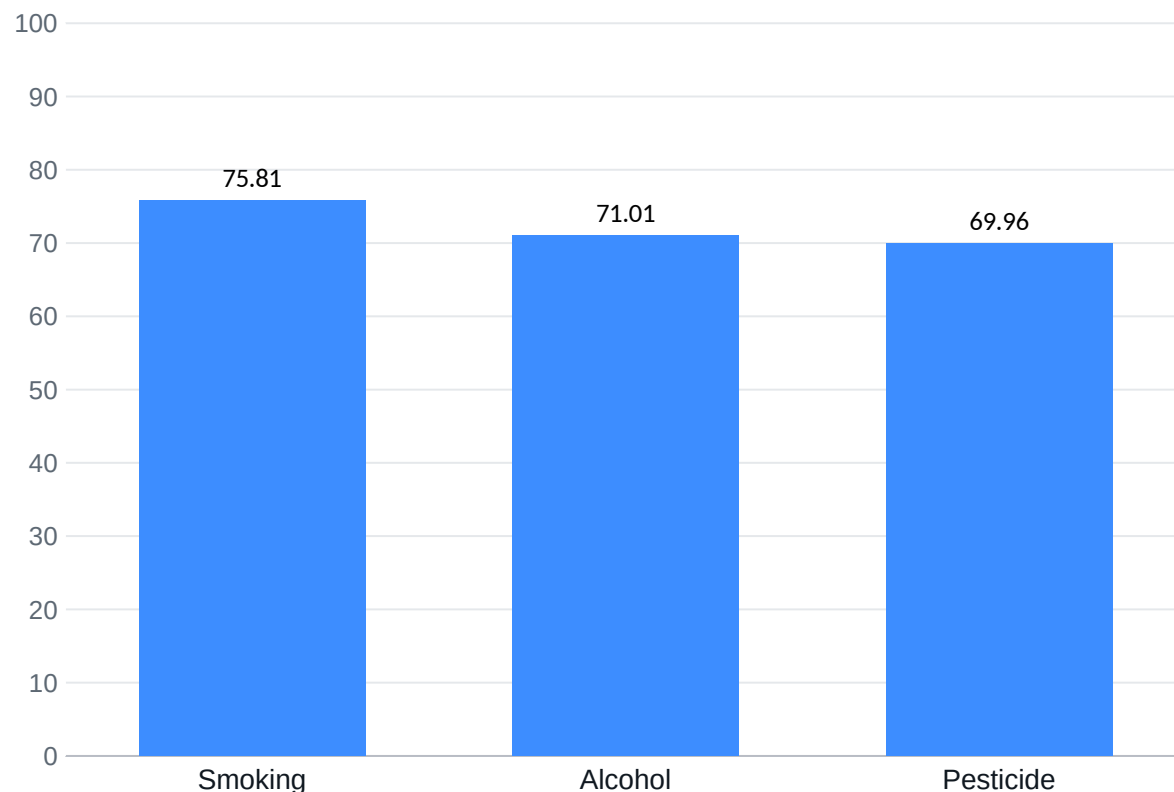
124 face biopsies - highest by body region

Sanitation result

Poor-access records had 22.3% malignant diagnoses versus 61.8% with good access. This dataset does not support poor sanitation as a severity driver.

Access: 782 patients lacked piped water; 815 lacked sewage systems.

Smoking and drinking mark a high-risk subgroup



82.1%

malignant among patients
who both smoke and drink

45.2%

of smokers also
consume alcohol

Interpret carefully

Only 62 patients smoke and 28 both smoke and drink. The pattern is strong, but the sample is small and the analysis does not prove causation.

Four findings should shape DORI's prevention strategy

01

Older adults create the greatest screening demand

Ages 60-79 represent 42.3% of all records, making age-targeted recall a practical first step.

02

Symptoms and lesion size can sharpen escalation rules

Itching and elevation are common; melanoma has the largest average diameter and needs rapid review.

03

Face lesions concentrate malignant and biopsy workloads

The face records 102 malignant diagnoses and 124 biopsies, supporting site-specific vigilance.

04

Lifestyle clustering identifies a focused prevention group

Smoking plus alcohol is linked to an 82.1% malignant rate, but the small sample requires validation.

Target the strongest signals to improve early detection

01

PRIORITISE

Recall patients aged 60-79 and anyone with previous skin-cancer history for routine lesion review.

02

TRIAGE

Fast-track growing, itching, elevated or changing lesions, especially large lesions and facial sites.

03

PREVENT

Deliver combined smoking-and-alcohol counselling and targeted pesticide-safety education.

04

VALIDATE

Track diagnosis stage, lesion size and follow-up outcomes; adjust for age and region before causal claims.

Success measure: earlier-stage confirmation, shorter time to biopsy, and fewer advanced malignant presentations.